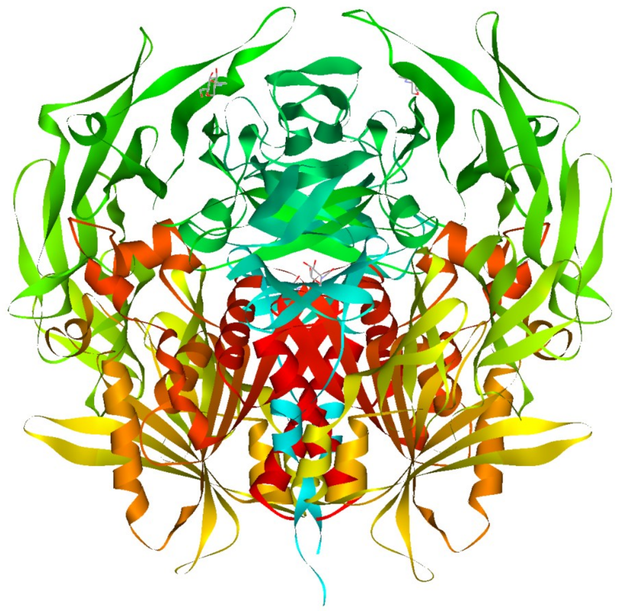
Virtual Screening, Synthesis, and Biological Evaluation of Some Carbohydrazide Derivatives as Potential DPP-IV Inhibitors

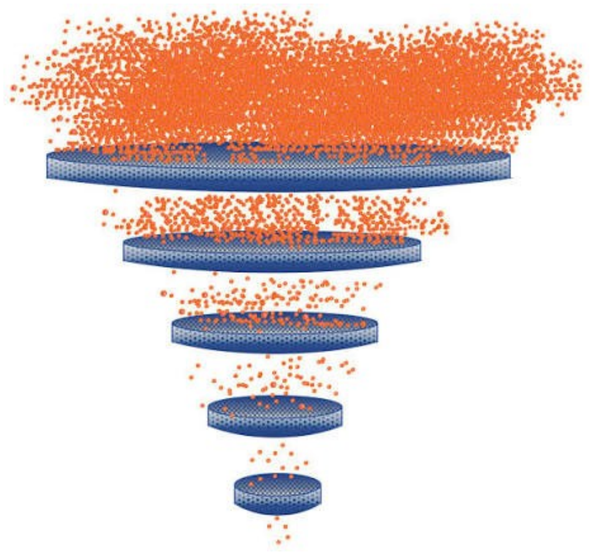
Jadhav PB, Jadhav SB, Zehravi M, Mubarak MS, Islam F, Jeandet P, Khan SL, Hossain N, Rashid S, Ming LC, et al. Virtual Screening, Synthesis, and Biological Evaluation of Some Carbohydrazide Derivatives as Potential DPP-IV Inhibitors. Molecules. 2023; 28(1):149. https://doi.org/10.3390/molecules28010149

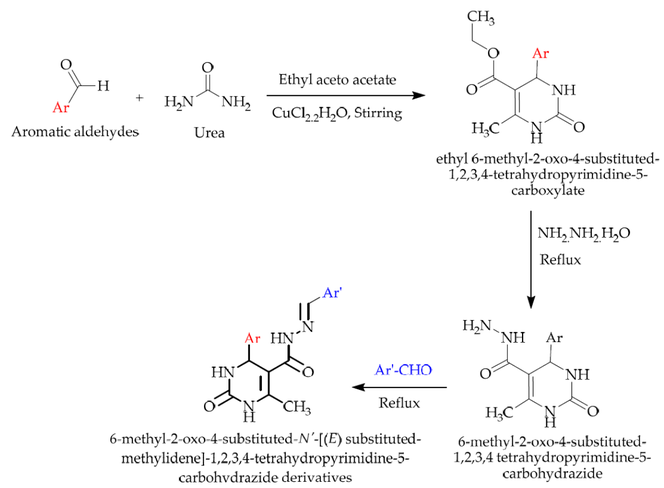
Context and target:
Target: Dipeptidyl peptidase-4 (DPP-IV)
Relevance: GLP-1/GIP regulation
(Glycemic control without hypoglycemia)
Key therapeutic target in Type 2 Diabetes

VS strategies:
Designed Library: 20 derivatives
Filter 1: Lipinski & Veber rules
Filter 2: ADMET & Toxicity (ProTox-II)
Filter 3: Docking (AutoDock Vina)

Synthesis:
Biginelli multicomponent reaction
Hydrazide formation & Schiff base condensation
Spectral confirmation: FTIR, NMR (^1H, ^{13}C), MS



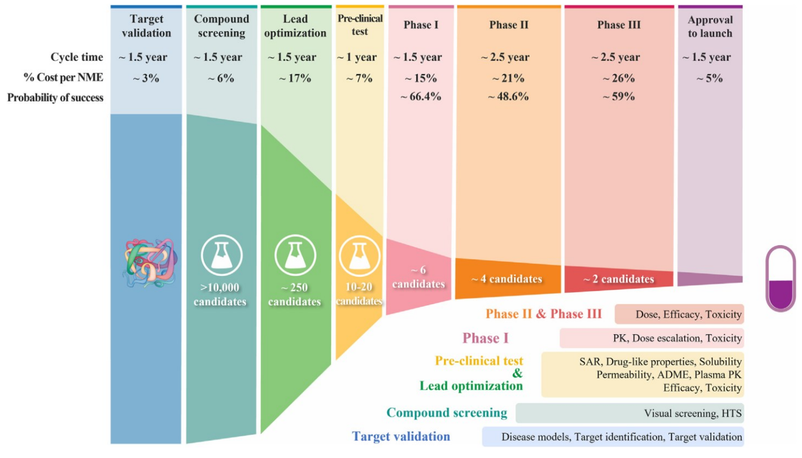


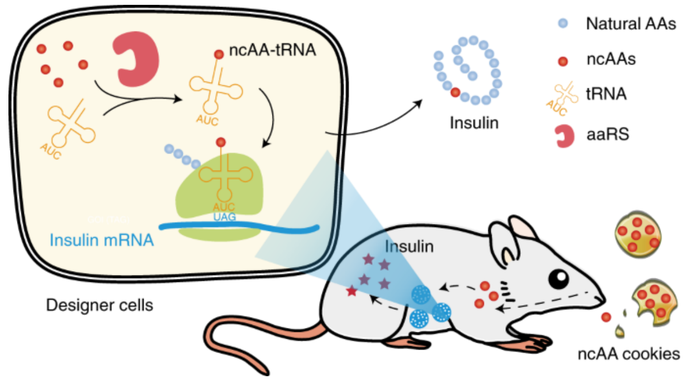


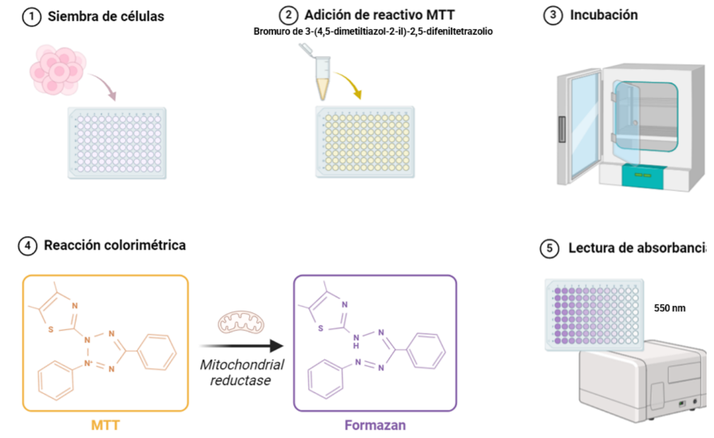
Impact and significance:
Drastic reduction of R&D cost and time
Identification of derivative 5b as an oral lead
Rational SAR established for novel antidiabetics

In Vivo Assays:
STZ-induced diabetic Wistar rats
Oral dose: 50 mg/kg over 21 days
Day 21 blood glucose: 157.33 mg/dL (5b)
vs 280.00 mg/dL (Diabetic control)

In Vitro Assays:
Fluorometric enzyme inhibition assay
Hits: 5b (IC_{50} = 28.13 μM), 4c
Standard: Sitagliptin (IC_{50} = 0.018 μM)







Glossary of technical terminology

ADMET Profile

ADMET is the acronym for Absorption, Distribution, Metabolism, Excretion, and Toxicity—the pharmacokinetic properties that govern whether an active molecule can reach therapeutic targets in vivo without inducing unacceptable adverse effects.

ProTox-II

An open-access web server designed for in silico chemical toxicity prediction. It integrates machine learning classifiers trained on diverse experimental datasets, combining chemical similarity, functional fragments, and pharmacophore matching. It predicts acute oral toxicity (median lethal dose, LD_{50}), assigns hazard toxicity classes, and quantifies probabilities for target organ toxicities, such as hepatotoxicity, mutagenicity, and carcinogenicity.

Biginelli Multicomponent Reaction

A classical multicomponent organic reaction reported by Pietro Biginelli in 1893. It involves the one-pot cyclocondensation of an aromatic aldehyde, a β-ketoester (such as ethyl acetoacetate), and urea or thiourea in the presence of an acid or Lewis acid catalyst $CuCl_2.2H_2O$.

Streptozotocin (STZ)-Induced Type 2 Diabetes Model

A widely utilized experimental preclinical rodent model used to simulate human T2DM pathophysiology. Streptozotocin is a cytotoxic glucosamine-nitrosourea compound that selectively enters pancreatic β-cells via the GLUT2 glucose transporter, inducing DNA alkylation, oxidative stress, and necrosis. When administered after establishing dietary insulin resistance via a high-fat diet, it causes mild-to-moderate pancreatic dysfunction that mirrors late-stage Type 2 diabetes without causing total β- cell ablation.

Half-Maximal Inhibitory Concentration (IC_{50})

A standard quantitative pharmacological metric indicating the concentration of a chemical inhibitor required to suppress 50% of an enzyme's maximum biological activity in vitro. Lower IC_{50} values reflect higher inhibitory potency. In this study, IC_{50} was derived from dose-response inhibition curves against human DPP-IV.